a Input
Unworn training teeth (EDJ series: 17)
Worn scan to reconstruct
TEST1, level 5 (Molnar stage 5)
b Stage 1: dense correspondence (every tooth)
1 Sample
2 Normalise
3 Rigid alignment (ICP)
4 Deformation (CPD)
template
worn tooth
Points drawn uniformly on the surface: 100,000 per tooth (ULM series: 10,000)
Centred, scaled to unit size, rotated onto principal axes; 24 axis rotations tested
Aligned to the template (grey), the most central training tooth
Template warped onto the tooth; equal colour = same point index, so every tooth shares one ordering
one shared point order
c Stage 2: reconstruction (three alternative methods)
2a Global SSM
Shape model: first mode of variation
−2σ
mean
+2σ
Fit to the points that survive
observed (Ω)
estimated missing (15% proxy)
$\mathbf{b} = (\mathbf{V}^\top \mathbf{V} + \lambda \Lambda^{-1})^{-1} \mathbf{V}^\top (\mathbf{x}_\Omega - \mu_\Omega)$
coefficients clipped to ±4σ
2b Neighborhood SSM
PC2
PC1
Worn tooth (star) among the training teeth (2 of 10 PCs shown); the nearest K within median + 2 IQR are kept (K = 5)
Model rebuilt from those neighbours, then fitted as in 2a
2c Non-rigid fit
Starts from the 2b output
confidence
0
1
Confidence falls to 0 where the scan is missing. Confident points steer a thin-plate-spline warp onto the scan, repeated 12 times, coarse to fine
Every output is mapped back to millimetres, meshed (screened Poisson), trimmed and smoothed. Tint: anatomy removed by wear.
d Evaluation against the original
Held-out unworn original, wear mask in red
Error of the 2c output after rigid ICP
Mask: points of the original more than 2 median spacings from the worn scan
0
0.15
0.3 mm
Three metric layers
Primary
worn region vs original
ranks methods
Secondary
full surface vs original
supporting
Tertiary
full surface vs worn scan
fidelity only
Distances in mm; only the primary layer ranks methods.